FIGHTING CANCER · EXPLAINABLE ARTIFICIAL INTELLIGENCE

Explainable multimodal AI for skin cancer diagnosis support from medical images

Smartphone photo + clinical data → a cancer probability that is measured, explained and delivered through a web platform

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PRESENTED AT ICAIS'26

0.947
AUC on the held-out test set (358 images, 197 unseen patients)
95% CI [0.918, 0.971]

96.8%
sensitivity at the triage threshold fixed in advance on development data
specificity 78.5% · calibration ECE 0.059

+0.051
AUC gained from the image over clinical data alone
95% CI [+0.012, +0.091] · P(Δ ≤ 0) = 0.004

$p = 1.8 \cdot 10^{-12}$
occlusion-map faithfulness: the highlighted regions truly matter
drop of 0.29 vs 0.01 for random regions

① Background and objective

Basal cell carcinoma (BCC), squamous cell carcinoma (SCC) and melanoma (MEL) are first spotted by eye, often far from a dermatologist. A triage aid is only useful if its decision is **measured** and its reasoning **verifiable**. Key prior work: image + clinical fusion [3], concept-based explanation [4], dermatology vision-language models [2, 5].

Objective: from a smartphone photo and 8 clinical variables, estimate the probability that a lesion is cancerous, explain this estimate, and deliver it through a platform usable by physicians.

- **Q1** Does a dermatology-specific vision-language encoder (DermLIP) outperform a generic encoder?
- **Q2** Does the image add information beyond the clinical data?
- **Q3** Are the generated explanations faithful to the model?

② Data and protocol

PAD-UFES-20 [1]: 2,298 smartphone photos, 1,373 patients, 6 diagnoses, with clinical metadata.

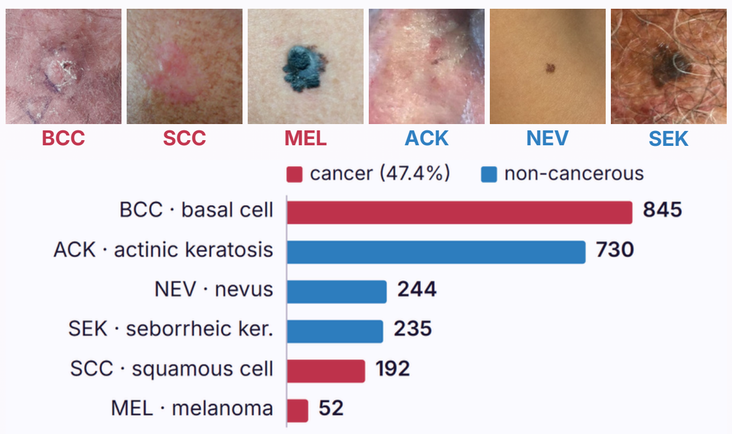


Fig. 1 — Images per diagnosis. Binary target: cancer (BCC, SCC, MEL) vs non-cancerous.

- **Patient-grouped split**: no patient appears in two sets. Test = 358 images (197 patients), used **only once**; development = 1,940 images with 5-fold cross-validation.
- Hyperparameters by nested cross-validation, model chosen by a pre-specified rule, CIs by patient bootstrap.

SHORTCUT DETECTED AND EXCLUDED

The patient profile (smoking, history…) is missing for 0% of cancers: its absence alone yields an AUC of 0.833 **without looking at the image**. Retained variables: age, location and 6 symptoms (itch, grew, pain, changed, bleeding, elevation).

③ Method

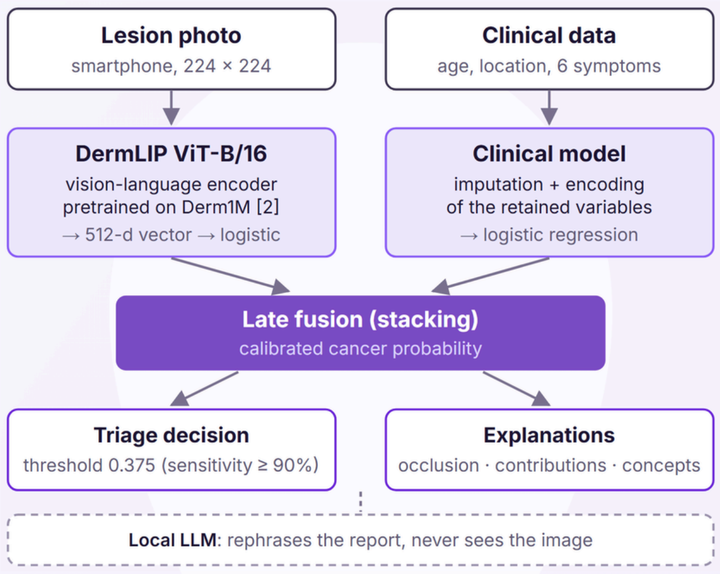


Fig. 2 — Decision pipeline. Only the trained and evaluated model produces the probability and the decision.

④ Results

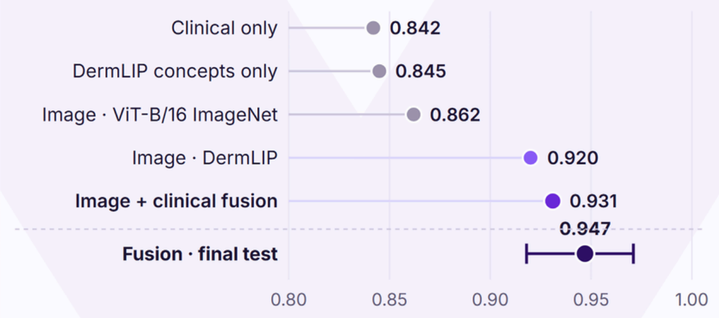


Fig. 3 — Cross-validated AUC (5 folds, development), then on the test set (bar: 95% CI). **Q1**: DermLIP +0.058 vs ImageNet (95% CI [+0.044, +0.072]). **Q2**: fusion beats clinical data alone.

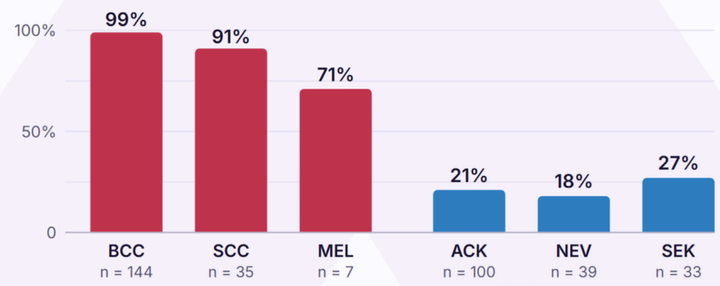


Fig. 4 — Share of lesions classified as "cancer" at the triage threshold, by diagnosis (test). Crimson: detected cancers; blue: false alarms.

THRESHOLD (SET ON DEV.)	SENSITIVITY	SPECIFICITY	BAL. ACCURACY
0.50	90.3%	83.7%	0.870
Youden (0.477)	91.4%	83.1%	0.873
Triage (0.375)	**96.8%**	**78.5%**	**0.876**

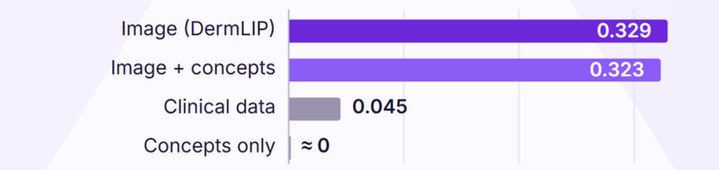


Fig. 5 — Permutation importance (AUC drop when a modality is shuffled): the decision relies primarily on the image.

⑤ Explainability

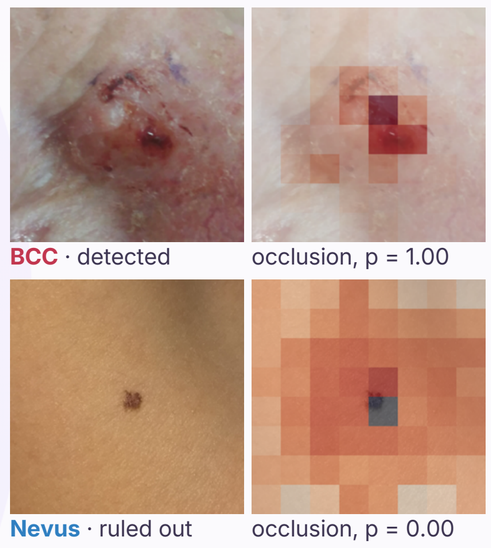


Fig. 6 — Occlusion maps on test images: each 8 × 8 patch is masked in turn; the darker the patch, the more the probability drops without it.

- **Occlusion**: where the model looked.
- **Contributions** of each factor (baseline Shapley values, exact for the linear part).
- Visual **concepts** measured by DermLIP (image-text similarity), for descriptive purposes.

Q3 FAITHFULNESS VERIFIED

Masking the highlighted regions drops the probability by **0.29** on average, vs **0.01** for random regions (Wilcoxon test, $p = 1.8 \cdot 10^{-12}$).

IDENTIFIED BIAS: INK MARKINGS

15.4% of melanomas carry surgical ink marks (≤ 5% elsewhere), and the model gives them 1.5 times more attention than their area warrants.

⑥ DermAI-XAI platform

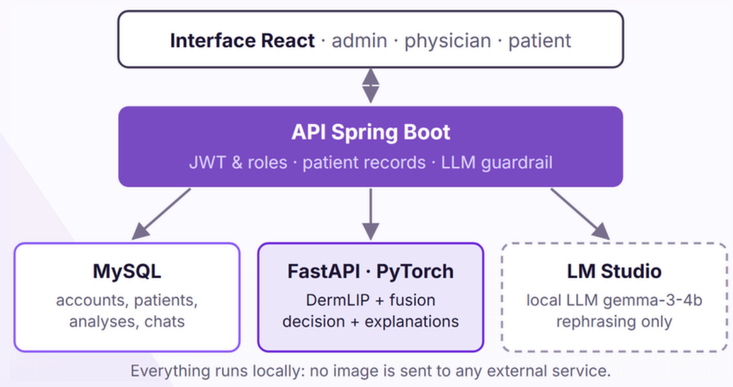


Fig. 7 — Web platform architecture.

- **Physician**: patient records, probability trend over time, physician- or patient-oriented report.
- **Chat**: text refines the symptoms; only a photo triggers the model. The LLM never sees the image, and a guardrail checks that it alters neither the probability nor the conclusion.
- **Administrator**: physician approval, roles, activity monitoring.

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⑦ Conclusion

- **Q1** The dermatology encoder DermLIP outperforms the generic encoder: **+0.058 AUC**.
- **Q2** The image adds **+0.051 AUC** beyond clinical data; test AUC 0.947.
- **Q3** Explanations are faithful ($p = 1.8 \cdot 10^{-12}$) and **revealed a real bias**.

⑧ Limitations and future work

- Few melanomas (52, of which 7 in test), few dark skin types (phototypes IV–VI), a single center (Brazil).
- External validation on Moroccan patients and mitigation of the ink bias.
- Prospective study with dermatologists. DermLIP: CC BY-NC-ND 4.0 license (research use).

References

1. Pacheco A.G.C. et al. PAD-UFES-20: a skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020.

2. Yan S. et al. Derm1M: a million-scale vision-language dataset aligned with clinical ontology knowledge for dermatology. ICCV, 2025.

3. Chakkarapani V. et al. Enhancing skin cancer detection with multimodal data integration: a combined approach using images and clinical notes. *SN Computer Science*, 6(1), 2025.

4. Patrício C. et al. A two-step concept-based approach for enhanced interpretability and trust in skin lesion diagnosis. *Comput. Struct. Biotechnol. J.*, 28:71–79, 2025.

5. Zeng W. et al. MM-Skin: enhancing dermatology vision-language model with an image-text dataset derived from textbooks. ACM Multimedia, 2025.

The decision comes from the trained and evaluated model; the LLM is used for presentation only.

Research tool: decision support only; it replaces neither clinical examination nor biopsy.